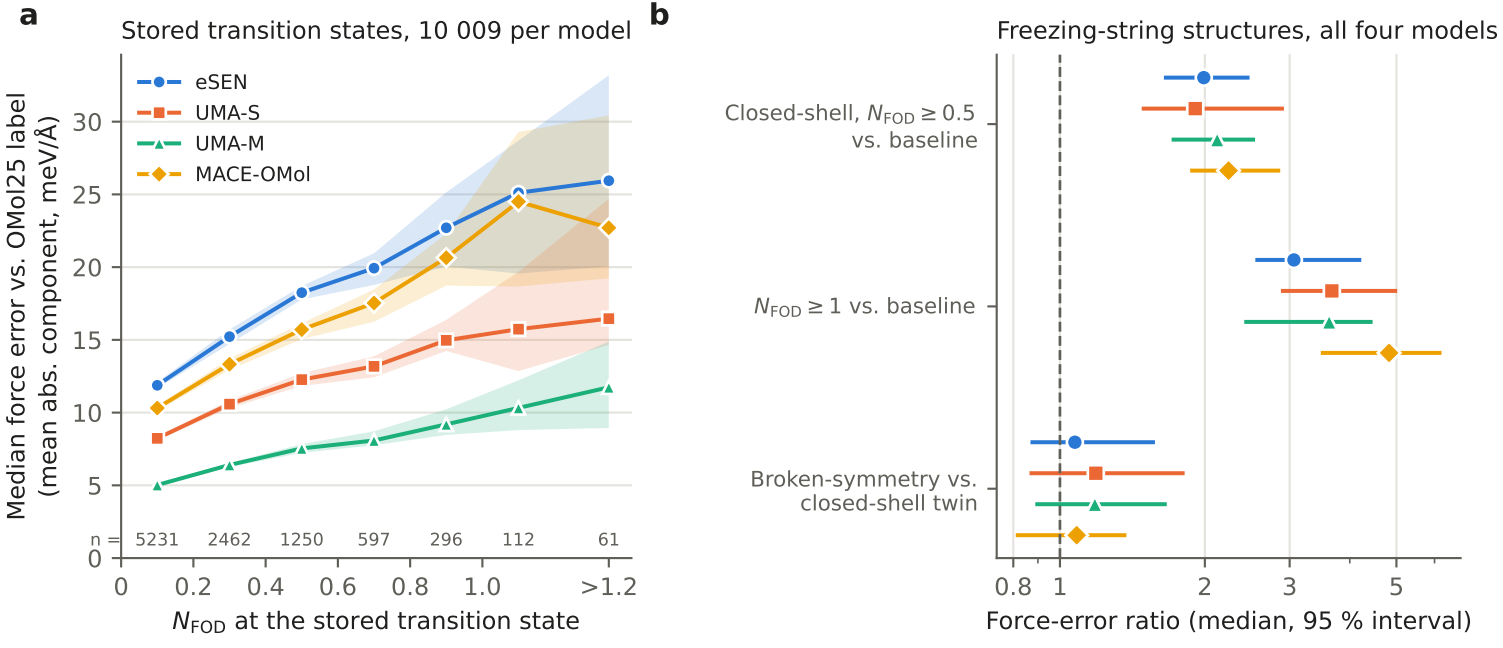
a
Stored transition states, 10 009 per model
eSEN
UMA-S
UMA-M
MACE-OMol
Median force error vs. OMol25 label
(mean abs. component, meV/Å)
30
25
20
15
10
5
0
n =
5231
2462
1250
597
296
112
61
0
0.2
0.4
0.6
0.8
1.0
>1.2
N_{FOD} at the stored transition state
b
Freezing-string structures, all four models
Closed-shell, $N_{FOD} \geq 0.5$
vs. baseline
$N_{FOD} \geq 1$ vs. baseline
Broken-symmetry vs.
closed-shell twin
0.8
1
2
3
5
Force-error ratio (median, 95 % interval)